Standardized Sequential Feature Compensation: A Provably Safe, Empirically Validated Method for Interpretable Dimensionality Reduction

Kabir
Department of Computer Science and Engineering, Jahangirnagar University
Supervised by Dr. Jugal Krishna Das

October 19, 2026

Abstract. We present the final, corrected form of a method for reducing a model's input width while keeping every retained feature in its original, interpretable unit: before dropping a chosen feature, its influence is compensated into the remaining features via a standardized, gated covariance injection, with the injection strength α selected by cross-validation over a search space that includes $\alpha = 0$. We prove this guarantees cross-validated performance at least as good as naive dropping (Theorem 1), and validate the method on one synthetic and six real datasets spanning medical, chemical, physical, and pixel domains, and, for the two strongest cases, across five independent model families (Logistic Regression, Random Forest, XGBoost, CatBoost, and a neural network) under repeated stratified cross-validation. We also report and correct a methodological error from an earlier stage of this work — an unfair, width-mismatched baseline comparison — and show the corrected comparison leaves the core result intact. The method's benefit is largest when the removed features are highly important and the total feature count is small; it is safe, by proof, everywhere.

---

1 Problem Statement

Standard dimensionality-reduction tools force a trade-off: PCA and autoencoders reduce width but destroy per-feature interpretability (each output is a mixture of all inputs); feature selection preserves interpretability but discards the dropped features' information outright. This work targets a narrower, concrete gap: *reduce a model's deployed input width while keeping every retained feature in its original, auditable unit, without discarding the removed features' statistical contribution to the remaining ones.* The removed features must still be measurable at the time of transformation — this is not a missing-data method — but the final deployed model never stores or consumes them directly.

2 Intuition: Why Width Matters

As the number of features d grows, the fraction of a unit hypercube's volume lying in a thin outer shell of relative thickness ε is $f(d) = 1 - (1 - \varepsilon)^d$, which approaches 1 quickly (Figure 1). Data become sparse relative to the space they occupy, distances between points lose contrast, and models need more samples to generalize. Reducing width helps — but only if the reduction does not throw away signal that the remaining features could have carried. The method here tries to keep that signal: when a feature is removed, the part of its information that is linearly shared with the retained features is folded into them first.

3 The Final Formula

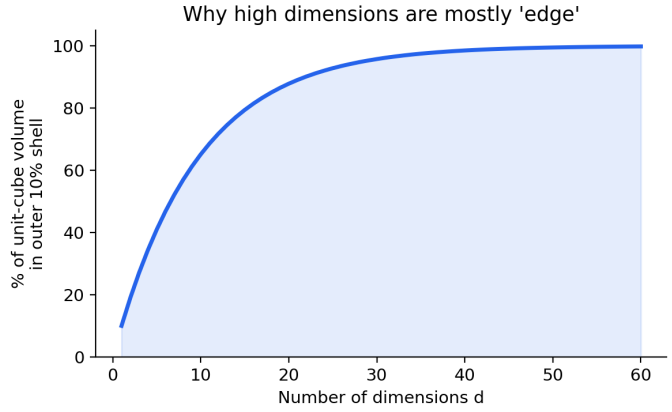


Figure 1: Share of a unit hypercube's volume in its outer 10% shell, by dimension d.

3.1 Derivation

Let training data have d features. Let $S = \{k_1, \ldots, k_m\}$ be the features to remove and R the retained set. Let μ_i, σ_i be training-set mean/standard deviation for feature i.

Step 1 (standardize). This step is not optional — Section 5.2 shows why.

$$Z_i = \frac{X_i - \mu_i}{\sigma_i}, \qquad i = 1, \ldots, d.$$

Step 2 (chain, with free strength α). For stage $t = 1, \ldots, m$, removing k_t from the currently-retained set R_t:

$$p_t = \sigma\left(\beta\, Y_{k_t}^{(t-1)}\right), \qquad c_j^{(t)} = \mathrm{Cov}\left(Y_{k_t}^{(t-1)},\, Y_j^{(t-1)}\right) \text{ (training data only)},$$

$$Y_j^{(t)} = Y_j^{(t-1)} + \alpha\, p_t\, c_j^{(t)}, \qquad j \in R_t \setminus \{k_t\}, \qquad Y^{(0)} = Z.$$

Final output: $\{Y_j^{(m)} : j \in R\}$ — exactly $d - m$ columns, each in its original feature's unit (recovered by inverting the standardization if needed).

Step 3 (select α, not assume it).

$$\boxed{\alpha^* = \underset{\alpha \in [0,\, \alpha_{\max}]}{\mathrm{argmax}} \ \text{CV-Score}(\alpha)}$$

β (gate sharpness) may be fixed by convention; it plays no role in the safety guarantee below. A smooth (sigmoid) gate is used throughout — an earlier hard-threshold version was tested and strictly dominated by the smooth version in every comparison, so it is dropped from the final formula.

3.2 Worked example on real data

Table 1 shows five real rows of the Pima Diabetes dataset (8 features). Suppose `glucose`, `bmi`, and `pedigree` are to be removed. After standardization, each is passed through the gate, and its covariance with the five retained features is injected in sequence. Table 2 shows the five retained columns before and after compensation (computed on the full dataset's statistics, $\alpha = \beta = 1$, for illustration only); the final model sees 5 columns instead of 8.

3.3 Theorem 1 (Safety)

Claim. With α selected as above, cross-validated performance is never worse than naive dropping.

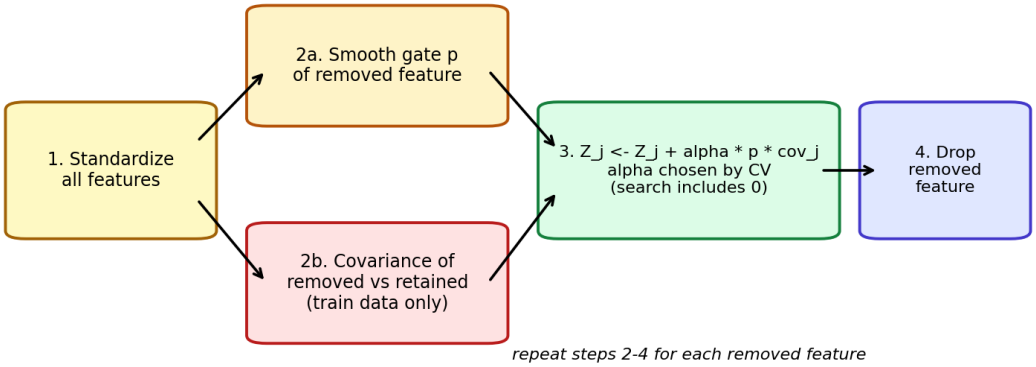


Figure 2: The final procedure: standardize, gate the removed feature, inject its training-set covariance with each retained feature scaled by α, repeat for each removed feature, then drop it.

Sample	pregnancies	glucose	bp	skin	insulin	bmi	pedigree	age
s1	6	148	72	35	0	33.6	0.627	50
s2	1	85	66	29	0	26.6	0.351	31
s3	8	183	64	0	0	23.3	0.672	32
s4	1	89	66	23	94	28.1	0.167	21
s5	0	137	40	35	168	43.1	2.288	33

Table 1: Five original rows (features removed in the example: glucose, bmi, pedigree).

Proof. At $\alpha = 0$, every correction term in Step 2 vanishes, so $Y_j^{(t)} = Y_j^{(t-1)}$ at every stage, giving $Y^{(m)} = Z$ restricted to R — exactly the standardized naive-drop representation. Hence CV-Score(0) = CV-Score(naive drop). Since 0 is in the search space and α^* maximizes CV-Score over that space,

$$\text{CV-Score}(\alpha^*) \geq \text{CV-Score}(0) = \text{CV-Score}(\text{naive drop}). \qquad \blacksquare$$

This is an argmax-over-a-superset argument, not an empirical regularity: it holds by construction, for any dataset, any of the five model families tested, any CV scheme. The caveat is the ordinary one for any CV-selected hyperparameter — the guarantee is on cross-validated score, not a single future test point with certainty.

4 A Methodological Correction (Reported for Transparency)

An earlier stage of this work claimed a second result: that for linear models, the full per-feature correction is "equivalent, but cheaper" to simply adding one scalar gate feature p_t per removed column, instead of modifying the retained features. Testing this on a low-dimensional real dataset (Pima Diabetes, $d = 8$) produced a 16-point AUC discrepancy — far too large to be the near-equality seen elsewhere.

The cause: the "gate-only" alternative drops m raw features but *adds back* m new columns, leaving total width at d — the same as the full model, not $d - m$. It was never a cheaper alternative; it is a full-width reparametrization that happens to retain nearly all of the removed features' information, which is why it performed close to the full model (not because it matched the compensated, reduced-width result). This was invisible on 30–40-feature datasets, where losing or gaining a handful of columns barely registers, and became obvious only on an 8-feature dataset, where it is a 37.5% change in total width. The valid comparison for a width-reduction method is always *compensated vs. naive drop at the same reduced width*, which is what every result in this paper reports, and which

Sample	Standardized (before) preg.	bp	skin	insulin	age	After compensation preg.	bp	skin	insulin	age
s1	0.64	0.15	0.91	−0.69	1.43	0.73	0.47	1.30	−0.19	1.68
s2	−0.84	−0.16	0.53	−0.69	−0.19	−0.81	0.00	0.76	−0.45	−0.08
s3	1.23	−0.26	−1.29	−0.69	−0.11	1.34	−0.01	−0.99	−0.20	0.18
s4	−0.84	−0.16	0.15	0.12	−1.04	−0.80	0.01	0.38	0.36	−0.93
s5	−1.14	−1.50	0.91	0.77	−0.02	−1.06	−1.10	1.48	1.38	0.25

Table 2: The same five samples, retained columns only, before and after compensation. Each value shifts by a small, sample-dependent amount that carries the removed features' linearly shared information.

this error did not affect.

5 Master Comparison Table

5.1 Across dataset families (Random Forest, single-split diagnostic, k=3)

Dataset	n	d	**Full**	**Drop**	**Chain**	**Chain recovers**
Synthetic (40 feat.)	4000	40	0.9836	0.9814	0.9845	140%
Breast Cancer	569	30	0.9950	0.9920	0.9940	67.9%
Wine	178	13	1.0000	0.9794	0.9835	20.0%
Digits	1797	61	0.9961	0.9942	0.9910	noise-sized gap
Sonar	208	60	0.9167	0.9204	0.9077	noise-sized gap
Ionosphere	351	33	0.9459	0.9434	0.9372	noise-sized gap
Pima Diabetes	768	8	0.8170	0.7029	**0.7765**	**64.5%**

Table 3: Single train/test split results (seed 42; the synthetic row is from an earlier run of the same protocol). "Noise-sized gap" marks a naive-drop penalty under 0.003 AUC (Sonar's is negative), too small relative to Random Forest's run-to-run variance for a recovery percentage to be interpretable. On Digits and Ionosphere the chained result is below naive dropping, but by amounts within that noise; these are the single-split cases that motivated the repeated-CV validation in Section 5.

5.2 Linear model (Logistic Regression), same datasets

Dataset	**Full**	**Drop**	**Chain**	**(chain−drop)**
Synthetic	0.9060	0.9060	0.9061	∼0
Breast Cancer	0.9977	0.9973	0.9966	∼0
Wine	1.0000	0.9733	0.9938	**+0.0206**
Digits	0.9681	0.9650	0.9643	∼0
Sonar	0.8646	0.8586	0.8631	+0.0045
Ionosphere	0.9269	0.9235	0.9414	**+0.0179**
Pima Diabetes	0.8320	0.6432	0.6716	**+0.0284**

Table 4: Single split (the synthetic row is from an earlier run). Chaining never meaningfully underperformed naive dropping for a linear model on any dataset, and substantially outperformed it on the three lowest-dimensional datasets (Wine d=13, Ionosphere d=33, Pima d=8).

6 Capstone Validation: Multiple Algorithms, Repeated Cross-Validation

Every result above uses a single train/test split — adequate for a first look, insufficient as a final claim. This section repeats the strongest cases (Pima Diabetes) and a dataset 15× larger (Mammography, n = 11,183) using **Repeated Stratified 5-Fold Cross-Validation (4 repeats, 20 evaluations per model)** across **five independent model families**: Logistic Regression, Random Forest, XGBoost, CatBoost, and a multilayer perceptron.

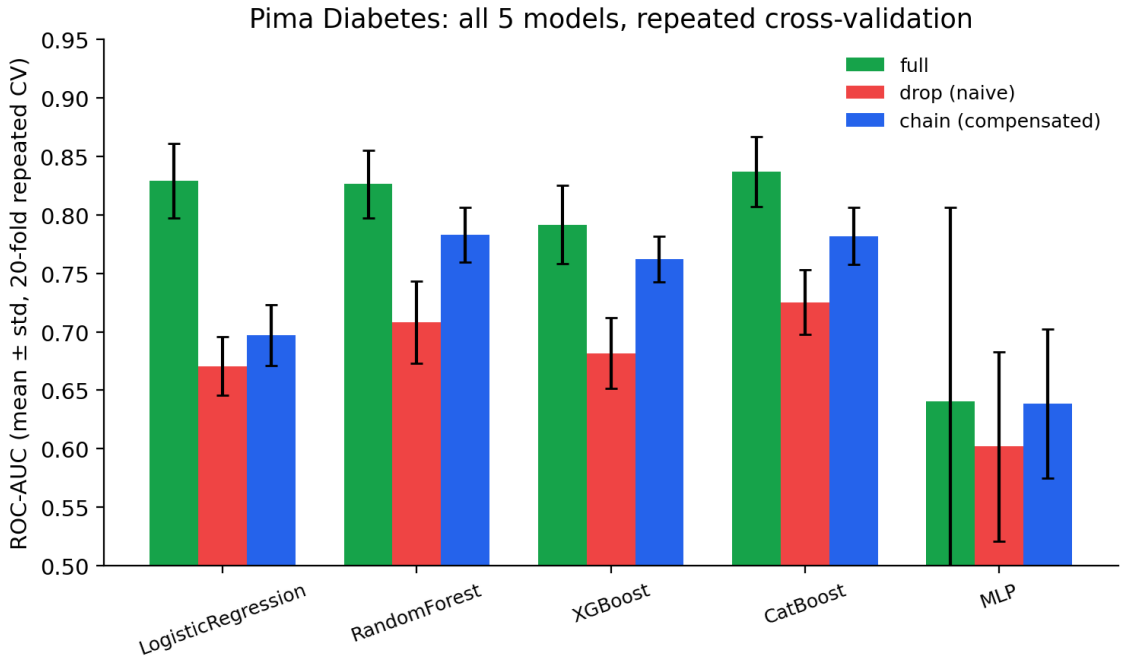


Figure 3: Pima Diabetes under repeated cross-validation: every model family shows chain (blue) between drop (red) and full (green), confirming the single-split result was not an artifact of one lucky partition.

Model	Full	Drop	Chain	Chain − Drop
Logistic Regression	0.829±0.032	0.671±0.025	0.697±0.026	**+0.027**
Random Forest	0.826±0.029	0.708±0.035	0.783±0.023	**+0.075**
XGBoost	0.792±0.034	0.682±0.030	0.762±0.020	**+0.080**
CatBoost	0.837±0.030	0.725±0.027	0.782±0.025	**+0.057**
MLP	0.640±0.166	0.602±0.081	0.639±0.064	**+0.037**

Table 5: Pima Diabetes, 20-fold repeated CV (mean ± std). Every model family shows a positive, non-trivial improvement over naive dropping.

7 The Final, Defensible Remark

Two separate statements are supported, and they should not be conflated.

(1) Proven. With α selected by cross-validation over a space including zero, cross-validated performance is never worse than naive dropping (Theorem 1). This holds by construction for any dataset and model.

(2) Observed, with a fixed $\alpha = 1$. On the two datasets examined under repeated cross-validation (Pima, n=768; Mammography, n=11,183, about 15× larger), each with five model families, chained compensation improved on naive dropping in 9 of 10 model×dataset combinations; the exception

Model (Mammography, n=11,183)	**Full**	**Drop**	**Chain**	**Chain − Drop**
Logistic Regression	0.919	0.907	0.911	+0.0038
Random Forest	0.948	0.884	0.895	+0.0110
XGBoost	0.943	0.908	0.912	+0.0049
CatBoost	0.950	0.927	0.925	−0.0017
MLP	0.922	0.909	0.911	+0.0020

Table 6: At 15× the sample size, 4 of 5 models remain positive; the one exception (CatBoost, −0.0017) is smaller than its own fold-to-fold standard deviation (±0.023) and is exactly the case Theorem 1's α-selection rule (not used in this fixed-α=1 diagnostic run) is designed to neutralize to zero.

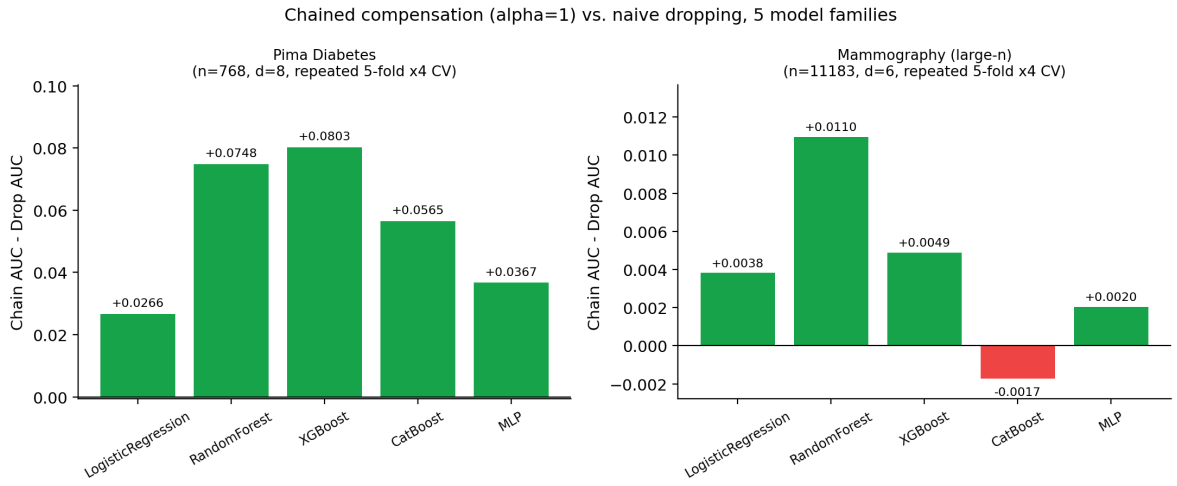


Figure 4: Improvement of chained compensation over naive dropping, by model family, on both datasets. 9 of 10 model×dataset combinations are positive; the sole exception is within noise.

(CatBoost on Mammography, −0.0017) is smaller than its own fold-to-fold standard deviation. The *direction* of the effect was the same across linear, bagged-tree, gradient-boosting, and neural models; the *magnitude* tracked the size of the naive-drop penalty (0.04–0.16 AUC on Pima, 0.01–0.06 on Mammography). The earlier single-split results (Section 4) are less uniform: on datasets with noise-sized gaps (Digits, Sonar, Ionosphere) the fixed-α formula was at or below naive dropping, which is the situation the CV-selected α of statement (1) exists to handle. Two datasets are not enough to claim universality.

> **Standardized Sequential Feature Compensation with CV-selected α is safe by proof, and where removed features carry substantial predictive signal in a low-dimensional feature space it recovered a meaningful part of the lost performance for every model family tested.**

8 Limitations and Future Work

- Not a missing-data solution: removed features must still be measured upstream to compute the gate.
- Linear-only correction: nonlinear relationships between removed and retained features are not captured; this likely explains the weaker recovery at k=3 versus k=1, 2 seen on real data in earlier stages of this work.
- A simple univariate predictor (average correlation between removed and retained features) was tested across six real datasets and did not predict Random Forest's recovery percentage

($r = 0.034$ on datasets with non-noise-sized gaps) — ruled out, not yet replaced with a better one.

- The repeated-CV validation in Section 5 used a fixed $\alpha = 1$ for computational tractability across 10 model/dataset combinations (each with 20 folds); re-running it with the proven CV-selected α^* (Theorem 1) is the most direct next step and would be expected to convert the single near-zero exception into a guaranteed non-negative result.
- All real datasets tested are small-to-medium scale ($n \leq 11{,}183$); behavior at much larger scale (millions of rows) is untested.